

of these giants are expected to be part of your sample, and your average is likely to be high. In other words, the largest observation cannot be too far away from the average. The average will always contain both kinds, giants and dwarves, so that neither should be too rare—unless you get a megagiant or a microdwarf on very rare occasion. This would be Mediocristan with a large unit of deviation.

Note once again the following principle: the rarer the event, the higher the error in our estimation of its probability—even when using the Gaussian.

Let me show you how the Gaussian bell curve sucks randomness out of life—which is why it is popular. We like it because it allows certainties! How? Through averaging, as I will discuss next.

How Coffee Drinking Can Be Safe

Recall from the Mediocristan discussion in [Chapter 3](#) that no single observation will impact your total. This property will be more and more significant as your population increases in size. The averages will become more and more stable, to the point where all samples will look alike.

I've had plenty of cups of coffee in my life (it's my principal addiction). I have never seen a cup jump two feet from my desk, nor has coffee spilled spontaneously on this manuscript without intervention (even in Russia). Indeed, it will take more than a mild coffee addiction to witness such an event; it would require more lifetimes than is perhaps conceivable—the odds are so small, one in so many zeroes, that it would be impossible for me to write them down in my free time.

Yet physical reality makes it possible for my coffee cup to jump—very unlikely, but possible. Particles jump around all the time. How come the coffee cup, itself composed of jumping particles, does not? The reason is, simply, that for the cup to jump would require that all of the particles jump in the *same* direction, and do so in lockstep several times in a row (with a compensating move of the table in the opposite direction). All several trillion particles in my coffee cup are not going to jump in the same direction; this is not going to happen in the lifetime of this universe. So I can safely put the coffee cup on the edge of my writing table and worry about more serious sources of uncertainty.

FIGURE 7: How the Law of Large Numbers Works
